1. Clinical Question & Core Hypothesis

Clinical Question

In muscle-invasive disease treated with pembrolizumab plus cytotoxic chemotherapy, why do some tumors fail to convert chemo-induced tumor cell death into effective T cell–mediated clearance? 2) Tier 1 paths used: Core Path 1 (clinical testing of pembrolizumab in muscle-invasive patients) and Core Path 2 (synergy of pembrolizumab + chemotherapy leading to cell death) provide the treatment context and mechanistic expectation that chemo can create immunogenic tumor cell death that PD-1 blockade should exploit. I built on Core Path 3 (KRAS mutational context) as a tumor-intrinsic modifier that can reprogram death/immune signaling. 3) Tier 2 support: links between KRAS signaling and cell-death programs (KRAS → apoptotic/necrotic bias), the PD-1/PD-L1 axis (PDCD1/CD274) as the immunotherapy target, and cisplatin's DNA-damage/immunogenic effects support a mechanism where KRAS alters chemo-induced immunogenicity and PD-L1/T cell dynamics. 4) Endpoint choice: I selected therapy_response (pembrolizumab + chemotherapy response) because the core paths are therapeutic (trial + combo therapy) and the mechanistic question asks why response is variable; therapy_response is the most proximate, testable clinical readout for that puzzle. 5) Deliberate exclusions: I excluded broad, non-specific ideas like "drug-induced new mutations" as primary modifiers because they are diffuse and not directly testable in the initial focused hypothesis; I also withheld tissue-context claims (e.g., colon-specific KRAS biology) from the core hypothesis because the ontologist flagged cross-cancer context as uncertain and because the hypothesis targets muscle-invasive disease specifically?

Working Hypothesis

Activating KRAS signaling in muscle-invasive bladder cancer dampens chemo-induced immunogenic cell death and impairs T cell–mediated anti-tumor immunity, thereby reducing the therapeutic efficacy of pembrolizumab in combination with platinum-based chemotherapy. Specifically, KRAS-mutant tumors will exhibit (a) attenuated IFN-γ–driven upregulation of PD-L1 (CD274) and diminished antigen-presentation signaling after chemotherapy, (b) lower intratumoral TIL density following chemo+ipi (immune checkpoint blockade) treatment, and (c) inferior therapy_response compared with KRAS-wildtype tumors that possess high baseline or chemo-induced TILs. Secondary modifiers such as PDCD1 (PD-1) expression on T cells and downstream KRAS/MAPK signaling will modulate the magnitude of this effect. Testable predictions: (1) KRAS-mutant, TIL-low tumors will show the least response to pembrolizumab+chemotherapy; (2) post-treatment biopsies will reveal reduced TIL density and lower CD274 induction in KRAS-mutant tumors; (3) pharmacologic or genetic dampening of KRAS/MAPK signaling or restoration of antigen presentation will enhance chemo-immunotherapy–driven TIL recruitment and improve therapy_response in KRAS-mutant models.

Element	Value
Imaging feature	Tumor-Infiltrating Lymphocyte (TIL) density
Anchor gene(s)	KRAS, CD274
Primary endpoint	therapy_response

2. Why This Gene — Not Others

Primary hypothesis gene(s)

KRAS: KRAS signaling and cell-death programs (KRAS → apoptotic/necrotic bias), the PD-1/PD-L1 axis (PDCD1/CD274) as the immunotherapy target, and cisplatin's DNA-damage/immunogenic effects support a mechanism where KRAS alters chemo-induced immunogenicity and PD-L1/T cell dynamics.

CD274: CD274) as the immunotherapy target, and cisplatin's DNA-damage/immunogenic effects support a mechanism where KRAS alters chemo-induced immunogenicity and PD-L1/T cell dynamics.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core
Secondary	PDCD1	Supporting modifier from knowledge graph
Secondary	MAPK pathway effectors / KRAS downstream signaling	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: upregulation is often IFN-γ–driven and may reflect an active immune microenvironment; however, in KRAS-mutant contexts, impaired antigen presentation and T cell trafficking can decouple PD-L1 expression from effective T cell–mediated killing, leading to poor response to pembrolizumab-containing regimens.

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Novel mechanistic hypothesis derived from knowledge graph analysis.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- PDCD1: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- MAPK pathway effectors / KRAS downstream signaling: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.